

Research papers

Integrating deep learning, satellite image processing, and spatial-temporal analysis for urban flood prediction

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ABSTRACT

Urban flooding is escalating worldwide due to the increasing impervious surfaces from urban developments and frequency of extreme rainfall events by climate change. Traditional flood extent prediction and mapping methods based on physical-based hydrological principles often face limitations due to model complexity and computational burden. In response to these challenges, there has been a notable shift toward satellite image processing and Artificial Intelligence (AI) based approaches, such as Deep Learning (DL) models, including architectures like Convolutional Neural Networks (CNN). The objective of this research is to predict near real-time (NRT) flood extents within urban areas. This research integrated CNN (U-Net) with Sentinel-1 satellite imagery, Digital Elevation Model (DEM), hydrologic soil group (HSG), imperviousness, and rainfall data to create a flood extent prediction model. To detect flooded areas, a binary raster map was created using calibrated backscatter values derived from the VV (vertical transmit and vertical receive) polarization mode of Sentinel-1 imagery, which was highlighted as having a significant impact on backscatter behavior and prediction results. Application of the model was demonstrated in urban areas of Miami-Dade County, Florida. The results demonstrated the capability of the model to provide rapid and accurate flood extent predictions at a spatial resolution of 10 m, with an overall accuracy of 97.05 %, F-1 Score of 92.49 %, and AUC of 93 % in the study area. The U-Net model's flood predictions were compared with historical floodplain data and then using GIS overlay analysis, resulting in a Ground Truth Index of 84.05 % that shows the accuracy of the model in identifying flooded areas. The research incorporated crucial flood-influencing data (including rainfall) to the flood extent prediction models and expanded the focus models beyond major rainfall events only to encompass a wider range of flood events. The presented NRT flood extent mapping model has a broad range of applicability, including, but not limited to, the continuous monitoring of flood events and their potential impacts on civil infrastructure assets (e.g., construction, operation, and maintenance of roads and bridges), early warning systems for timely evacuation and preparedness measures, and insurance risk assessment.

1. Introduction

Urban flooding, frequently triggered by intense rainfall events (pluvial flooding) within urban areas, is a widespread issue in the United States (US), impacting approximately 99 % of counties across the country (FEMA, 2020). Climate change has intensified extreme rainfall in urban areas, escalating pluvial flooding risks (Masson-Delmotte et al., 2021; Hathaway et al., 2024). Urban expansion further worsens these risks by increasing impervious surfaces (Ebrahimian et al., 2016a; b). National Oceanic and Atmospheric Administration (2022) reports that urban floods average \$4.7 billion in damages per event, severely

affecting infrastructure and daily routines. However, mapping the flooded areas soon after the rainfall can cut these damages by 30 % (ESCAP and Warning, 2023; Global Center on Adaptation, 2022). Implementing Near Real-Time (NRT) flood mapping systems can provide timely and accurate mapping of flood extents and enable quicker responses and more effective resource allocation, reducing the impacts of urban flooding (Hu and Demir, 2021; Huang et al., 2018; Hostache et al., 2018; Wang et al., 2022; Toth et al., 2000).

Traditional flood prediction methods, such as physics-based hydrological models, often struggle with real-time (NRT) applications due to their demanding computational requirements, challenging calibration

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processes, and lengthy simulation times, as noted in various studies (Bermúdez et al., 2018; Liao et al., 2023; Zahura et al., 2020; Zhang et al., 2021; Mohamadzaz et al., 2024). In contrast, Artificial Intelligence (AI) techniques such as Machine Learning (ML), utilizing extensive satellite data including Sentinel-1 imagery, provide a promising alternative. These methods aim to deliver faster and more precise flood predictions crucial for effective emergency response and management (Andaryani et al., 2021; Brunton and Kutz, 2022; Chang et al., 2022; Hosseiny et al., 2020; McGovern et al., 2017; Mosavi et al., 2018; Riazi et al., 2023; Toth et al., 2000; Young, 2002; ESA, 2023). Convolutional Neural Networks (CNN), especially U-Net models that are known for their pixel-wise segmentation capabilities, have demonstrated high effectiveness in identifying flooded urban areas and enhancing the speed of emergency responses in flood-prone regions (Kabir et al., 2020; Hosseiny, 2021; Hoese et al., 2020; Xu et al., 2019; Liao et al., 2023).

NRT flood mapping has evolved into two main approaches: direct detection using varying polarizations of satellite imagery and the integration of ML with key flood data, marking significant progress in the field (Zhou et al., 2021; Nemni et al., 2020; Vongkusolkiet et al., 2023; Katiyar et al., 2021). However, challenges remain, particularly with the temporal resolution of satellite images, which can delay updates and affect the responsiveness of flood mapping systems (Shen et al., 2019; Sunkpho and Ootamakorn, 2011). Enhancing these systems involves improving the temporal resolution and ensuring faster updates to match the rapidly changing nature of flood events. Furthermore, while Flood Susceptibility Mapping (FSM) using ML and satellite imagery has shown potential in identifying flood-prone areas, its effectiveness for NRT flood mapping is limited without the integration of real-time data (Andaryani et al., 2021; Riazi et al., 2023; Yu et al., 2023; Ahmadlou et al., 2021). To improve NRT flood monitoring, advancements are necessary in data integration, such as incorporating live rainfall or soil moisture data, which would enhance the system's adaptability and improve the accuracy of flood extent maps. While AI has significantly advanced flood mapping, further enhancements are needed to meet the demands of NRT flood mapping and ensure timely and effective responses to flooding events.

The objective of this research is to predict near real-time (NRT) flood extents within urban areas. This research integrated CNN (U-Net) with Sentinel-1 satellite imagery, Digital Elevation Model (DEM), hydrologic soil group (HSG), imperviousness, and rainfall data to create a flood extent prediction model. It is important to clarify that this study is focused on NRT flood extent prediction (aiming at the immediate identification and delineation of flood extents soon after rainfall occurs) rather than forecasting with a lead time, which typically provides advance warning of potential flooding, allowing for early preparations and preventative measures. Existing flood prediction studies either rely on hydrological models that are complex and computationally expensive with limited temporal and spatial resolutions (e.g., limiting the model development to two or three major flood events) or data-driven models using limited data (e.g., only focusing on a few flood events and using limited number of satellite images). This study addresses the limitations in temporal resolution by utilizing Sentinel-1 imagery from a time span of 2014 to 2023. This extended timeframe allows for a more comprehensive analysis of urban flooding events over the years. Additionally, other studies focused on major flood events in certain parts of the study areas and then generalized their results for the entire region. In contrast, this study takes the entire study region and uses input data that covers the entire area. This distinction suggests that the results of the study are not limited to specific flood events or localized areas within the study region. Moreover, one common limitation observed in previous studies is either neglecting the incorporation of rainfall data altogether or relying solely on annual (year-average) rainfall data, even when dealing with major flood events. This limitation was addressed here by using daily rainfall data from multiple rain gages across the study region in the developed flood prediction model. The inclusion of other flood influencing data (DEM, HSG, and imperviousness) together with the rainfall

data further enhances the model's predictive capabilities, surpassing the limitations observed in studies that focused on single data to enhance the model's generalizability.

2. Materials and methods

2.1. Data

Three major data groups were used in this research: flood influencing data, satellite imagery, and flood event reports. Each of these groups was further categorized into subcategories, based on the study objective and specific conditions of the ROI. The data description and sources can be found in Table 1.

2.1.1. Flood influencing data

In the initial phases of data preparation, the primary approach involved harnessing flood influencing datasets. These datasets encompass a mix of spatial information and gauge measurements. The selection of flood influencing data (including Digital Elevation Model (DEM), Hydrologic Soil Group (HSG), imperviousness, and rainfall) was justified based on their well-established roles in hydrological processes and their impact on the likelihood and severity of flooding, which were highlighted in several studies in the literature (Anni et al., 2020; Stefanidis and Stathis, 2013; Shuster et al., 2005; Morita, 2014; Ebrahimian and Mohamadzaz, 2021). Table 1 provides a thorough overview of the spatial data and sources utilized in this study. Notably, DEM serves as the foundational layer from which the slope data was calculated. DEM provides a three-dimensional representation of the region's topography, and the slope is a direct computational derivative that quantifies the change in elevation between adjacent pixels. This relationship is crucial, as slope is a key variable affecting water movement and accumulation, which are vital to understanding and predicting flood extents. The derived slope data is an integral component of the spatial analysis for our flood prediction model. Fig. 1 visualizes the DEM for the ROI. The distribution of flood influencing data is presented in Fig. 2, picturing the spatial pattern of slope, imperviousness, HSG, and rainfall for the ROI. Daily rainfall maps were organized and generated using the Thiessen polygon method for the period of 2014 to 2023, aligning with corresponding satellite imagery. As an example, the daily rainfall map from September 15, 2015, is presented in Fig. 2-d.

2.1.2. Satellite imagery

This study collected Ground Range-Detected Sentinel-1 Synthetic Aperture Radar (SAR) at high resolution and high reliability (IWGRDH)

Table 1
The data description and sources.

Data	Resolution	Description	Resources
DEM	1.524 m (5 ft)	Representation of Earth surface elevation that shows the terrain or landscape (ft)	United States Geological Survey (USGS) https://www.usgs.gov/the-national-map-data-delivery/
HSG	30 m	Categorized soils into four groups (A-D) based on their ability to infiltrate water	USDA Web Soil Survey https://websoilsurvey.sc.egov.usda.gov/
Imperviousness	30 m	Impervious surfaces (Presented in each pixel as a value between 0 to 100 %).	National Land Cover Database (NLCD) https://www.mrlc.gov
Rainfall	N/A	Daily rainfall (mm)	National Weather Service (NWS) https://www.ncdc.noaa.gov/cdo-web/datasets/GHCND/stations/

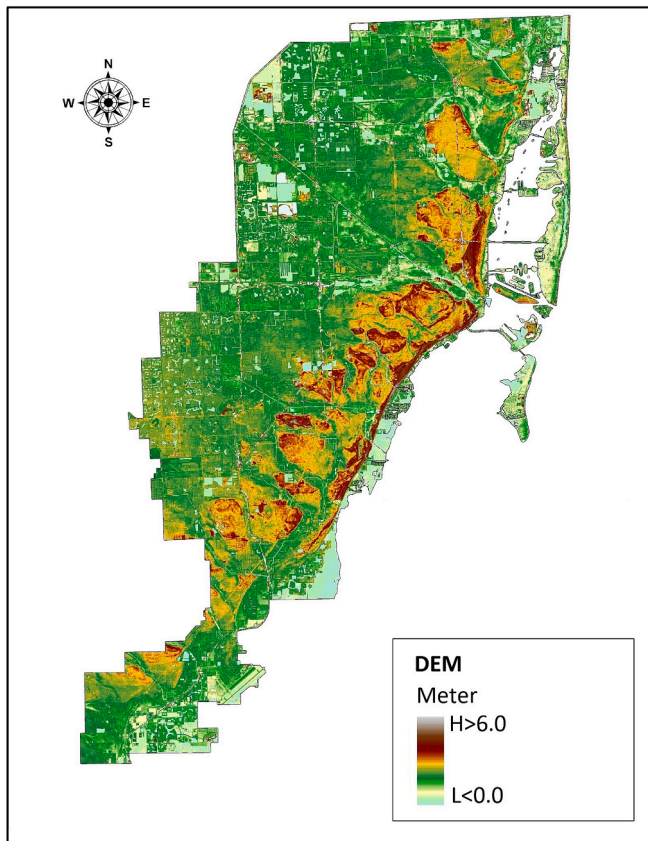


Fig. 1. LIDAR based topographic map, 1.5 m DEM of the ROI.

Level-1 (Copernicus, 2023) products for the time span of 2014–2023. A total number of 226 Sentinel-1 images were collected for this study. The temporal resolution of these images, crucial for monitoring changes over time, is consistent with the Sentinel-1 satellite's 6-day revisit cycle to any given point on Earth. These products contain backscatter amplitude data, characterized by a pixel spacing of 10 m (i.e., each pixel in the image represents a 10-m by 10-m square on the ground), a nominal spatial resolution of 20 m by 22 m, and a radiometric accuracy of 1 dB (3σ) (ESA, 2023). This high level of confidence means that when the radar sensor measures the intensity of radar signals reflected from the Earth's surface, the recorded values are expected to be very close to the true values, and one can be highly confident in the accuracy of these measurements (Torres et al., 2012). Future research can also use Google Earth Engine to access, pre-processed Sentinel-1 images. VV is the vertical transmit and vertical receive and VH is the vertical transmit and horizontal receive polarizations. The 20 m spatial resolution (10 m pixel spacing) Level-1 IWGRDH Sentinel-1 satellite images were provided with three colors (RGB) obtained from the Copernicus database. Copernicus services are readily accessible to users and come at no cost (<https://dataspace.copernicus.eu/>).

2.1.3. Flood event reports

Data collection for flood reports involved a multifaceted approach, drawing from both official sources and social media platforms. Firstly, the research investigated the 311-flood records (<https://www.miamigov.com/>), a database for local communities that allows residents and individuals to document and report flood-related incidents. This official channel served as a critical repository for firsthand accounts of flooding events within the ROI for the time 2014–2023. Second, ongoing flood events posted on Twitter obtained from the Institute for Environmental Studies (IVM) reports (<https://www.globalfloodmonitor.org/>), were extracted for the ROI from 2014 to 2023.

2.2. Methodology

This study encompasses multiple steps, from data preprocessing to implementing CNN for flood prediction using various data sources. A detailed breakdown of the methodology is presented in Fig. 3 and described as follows:

2.2.1. Raster map generating

The spatial flood influencing data (DEM, HSG, and imperviousness) for the ROI was properly preprocessed through the ESRI ArcGIS Pro 3.0.3 desktop GIS software (ESRI, 2023). This process ensured that the data was in a consistent format and spatial resolution, where each pixel (or grid cell) had a value representing the respective attribute.

Later, as an important step, the resolution and pixel spacing of the slope, HSG, and imperviousness were adjusted to be consistent across all three datasets, equal to 10 m (32.8 ft). This alignment was conducted to ensure that the same spatial resolution as the Sentinel-1 imagery was shared among all data layers. This is crucial because the CNN method requires inputs with uniform characteristics as emphasized by other studies (e.g., Kabir et al., 2020).

2.2.2. Rainfall data preprocessing

To accomplish the integration of daily rainfall records from gauge stations with the corresponding satellite images, first, the metadata of the Sentinel-1 satellite image was reviewed to determine the exact dates on which each image was captured. As a single Sentinel-1 satellite is mapping the Earth once every 12 days (in a single pass), the exact time of the day when the areas are imaged are known. Considering the temporal resolution for flood prediction is beneficial for effective inquiry into the rainfall data. Therefore, the review of Sentinel-1 imagery revealed alignment between daily accumulated rainfall data and the Sentinel-1 temporal resolution. This enabled direct association of daily rainfall amounts with corresponding Sentinel-1 imagery taken on the same day. Additionally, daily rainfall data is often readily available from meteorological stations, which makes it operationally feasible for flood prediction. Consequently, the tabular gauge rainfall data from the stations in the ROI were imported into the ArcGIS Pro. Out of several stations in the ROI, 7 rainfall stations were selected based on their historical rainfall data and alignment with the temporal resolution and period of this study from 2014 to 2023 (Table 2 and Fig. 4).

Estimating areal average of rainfall between rainfall stations was performed using the Thiessen Polygons method (De Silva et al., 2007). For each available date, this process was repeated and a standalone raster map for each date based on the corresponding rainfall measurements was generated using Python scripting (ArcPy).

2.2.3. Flood inventory map

Image thresholding was used to create a binary raster map for urban flooding detection in the ROI based on backscatter values from Sentinel-1 images, with low backscatter values indicating wet or flooded areas (Liang and Liu, 2020).

To prepare Sentinel-1 satellite images to apply into the CNN, a necessary preprocessing routine was performed using SNAP tool, a specialized tool for Sentinel data. The process within SNAP includes calibration for VV, speckle noise reduction to improve image quality, and terrain correction to accurately align images with geographic locations. Details of the preprocessing workflow for Sentinel-1 images are provided in the Supplementary Materials.

The Otsu method is the most common method for image thresholding, which is based on minimizing the variance on each of the classes (Di Baldassarre et al., 2009; Liu et al., 2020a; Dong et al., 2022; Liang and Liu, 2020; Liu et al., 2023; Yu et al., 2022; Cardille and M.A.Saah, 2023). Figure S2 illustrates the Otsu threshold and its deviation from the intersection point towards the center of the class with a larger variance (Xu et al., 2011). The choice of polarization affects accuracy of the flood detection, where VV polarization mode has been shown to detect and

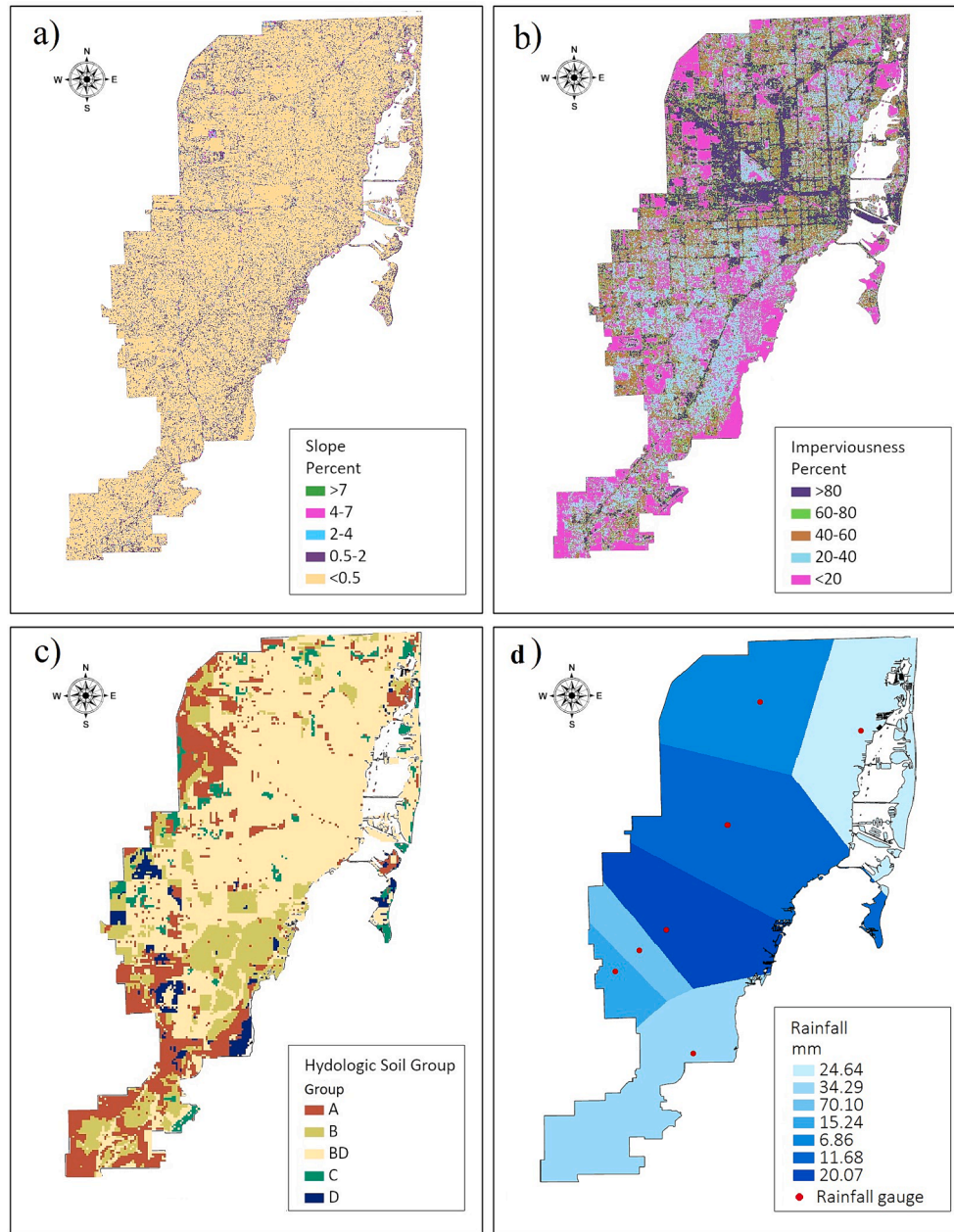


Fig. 2. Spatial distribution of flood influencing data in the ROI, a) slope, b) imperviousness, c) HSG, and d) average daily rainfall from Thiessen Polygons method for September 15, 2015.

highlight water bodies very effectively (Twele et al., 2016; Liang and Liu, 2020). Hence, the VV polarization mode was chosen to ensure clear and distinct results. This process resulted in creating a histogram that shows the distribution of pixel intensities in the image. The histogram displays one or more peaks of different magnitude considering the data. Backscatter values were converted to decibels (dB) using a logarithmic formula in Equation (1) (Richards, 2009) to make the data more interpretable. The SNAP tool was used to create a thresholding graph via the Graph Builder tool. The grayscale histogram $H(i)$ of an image, I , with an intensity level of i is defined by Equation (2). Once an optimal threshold value was determined using the Edge Otsu algorithm thresholding method, the subsequent action was to create a new image, essentially a binary representation where water and non-water areas were designated as 1 and 0, respectively (Equation (3)). This transformation was accomplished using Band Maths window calculations (McVittie, 2019; Cao et al., 2019).

$$\sigma_0 \text{ dB} = 10 \times \log_{10}(\sigma_0) \quad (1)$$

$$H(i) = \sum_{x=1}^m \sum_{y=1}^n [I(x, y) = i] \quad (2)$$

$$\text{Binary value} = \begin{cases} 1 & \text{if } \sigma_0 \text{ dB} < I(x, y) \\ 0 & \text{Otherwise} \end{cases} \quad (3)$$

where, $\sigma_0 \text{ dB}$ represents the backscattering image in decibels (dB), σ_0 is the original backscattering data (referred to as sigma + nought image), m and n represent the width and height of the image, and $\sigma_0 \text{ dB}$ refers to the backscatter values in the VV polarization mode. Details of the preprocessing workflow for Sentinel-1 images are provided in the [Supplementary Materials](#).

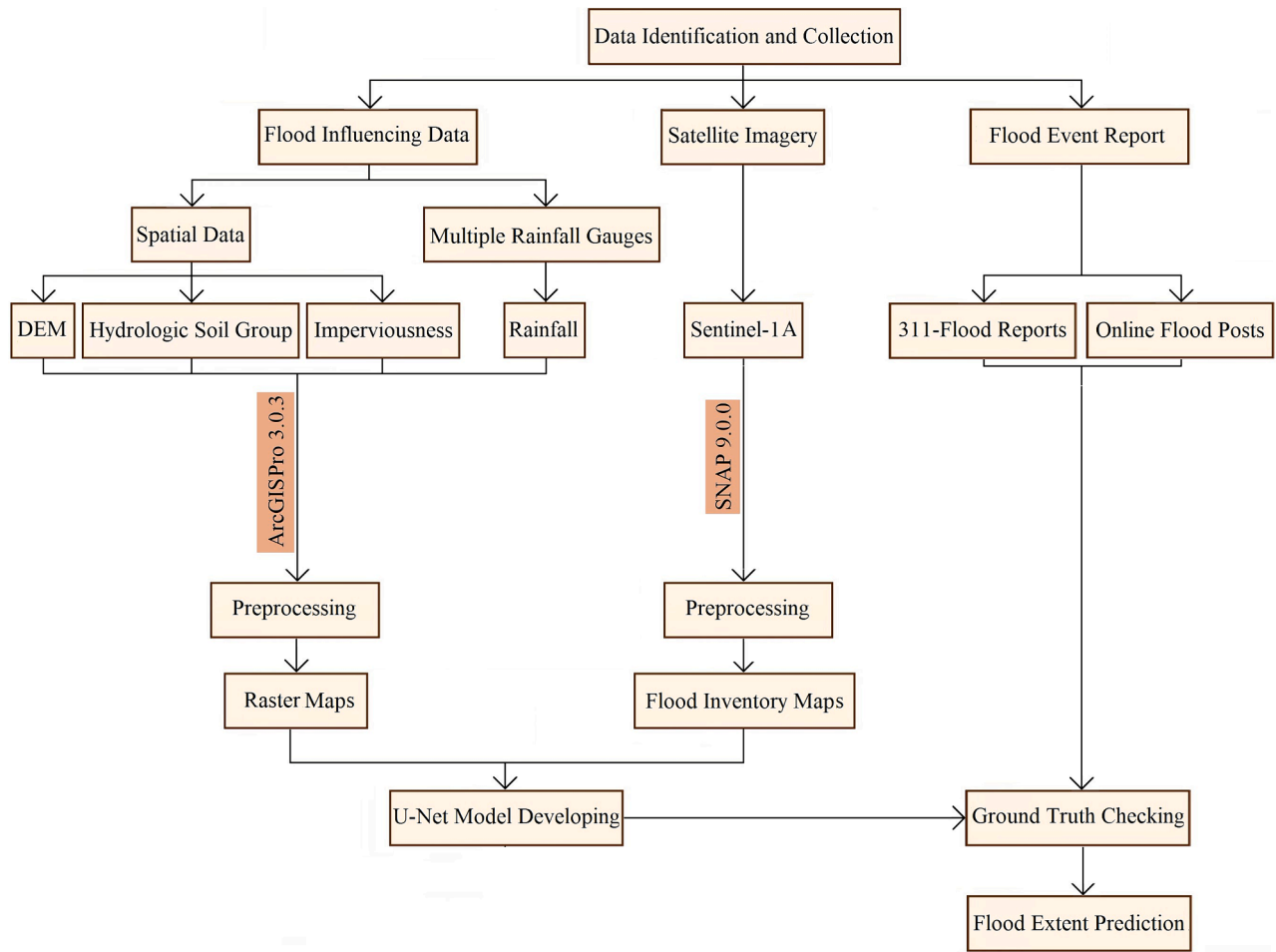


Fig. 3. Flowchart of the proposed methodology.

Table 2

Detailed information of rainfall stations in Miami-Dade County covering the Period of 2014 to 2023.

Station Name	Station ID	Latitude, Longitude	Coverage
Miami Kendall Tamiami EXEC Airport	USW00012888	25.642300, -80.434685	1998–2023
Cutler Ridge 1.5 SSW	US1FLMD0051	25.561026, -80.352631	2014–2023
Miami Opa Locka Airport	USW00012882	25.910189, -80.282936	1998–2023
Biscayne Park 0.3 E	US1FLMD0033	25.881929, -80.176849	2012–2023
North Miami Beach #2	USC00086315	25.950019, -80.215645	2001–2023
Miami International Airport	USW00012839	25.787990, -80.316925	1948–2023
Sunset 2.4 SW	US1FLMD0007	25.683458, -80.381470	2007–2023

2.2.4. U-Net model

CNN learns by adjusting its parameters (weights and biases) to minimize the difference between its predictions and the provided correct answers. In a two-dimensional image, the convolution process is defined as Equation (4) (Liu et al., 2020b; Sun and Scanlon, 2019).

$$X_{ij} = \sum_{u=1}^m \sum_{v=1}^n x_{i-u+1, j-v+1} \cdot f_{uv} + b \quad (4)$$

where x_{ij} represents the two-dimensional image array with indices

ranging from 1 to m for i and 1 to n for j . The convolution filter is denoted as f_{uv} with indices u ranging from 1 to m and v ranging from 1 to n , and b represents the bias (Hosseiny, 2021).

U-Net, built up by Li et al. (2023) upon the fully convolutional network with the goal of classifying each pixel in an image into categories of flooded or non-flooded was modified for the purpose of this study. Generally, the architecture follows the distinctive U-shaped design, consisting of both an encoder (contracting path) and a decoder (expansive path). In the encoding phase, the encoder is responsible for capturing contextual information and extracting high-level features from the input image. As visually depicted in Fig. 5, the number of feature maps increases progressively along the contracting path. Max pooling was employed to select the maximum values within specific windows, gradually reducing the spatial dimensions. The bottleneck, positioned at the center of the U-Net, serves to retain essential spatial information. Subsequently, the decoder undertakes the task of precise localization and generates the segmentation mask. Symmetrical to the contracting path, the decoder consists of several blocks, and each block is succeeded by an up-convolution or transposed convolution operation, effectively increasing the spatial resolution of the feature maps. This sequential flow of operations was designed to enhance the network's ability to learn and represent complex patterns for accurate pixel-level classification and segmentation tasks. One common activation function that allows learning and representing the complex pattern is Rectified Linear Unit (ReLU, Equation (5) (Hosseiny, 2021; Song et al., 2019)). This mathematical operation, applied after each convolution, sets all negative values to zero and allows positive values to pass through unchanged (Wu, 2017).

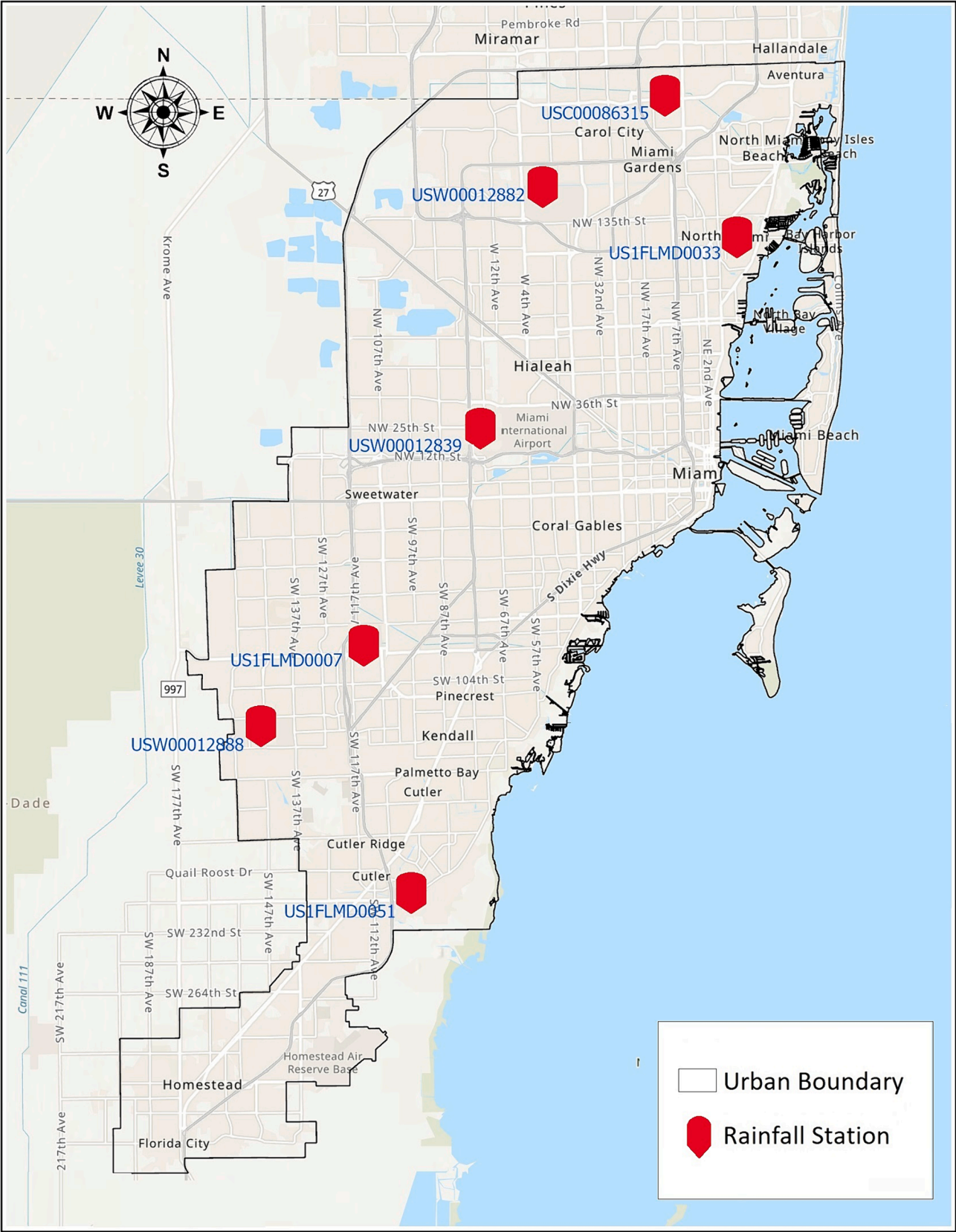


Fig. 4. Rainfall gauge stations' locations in the ROI.

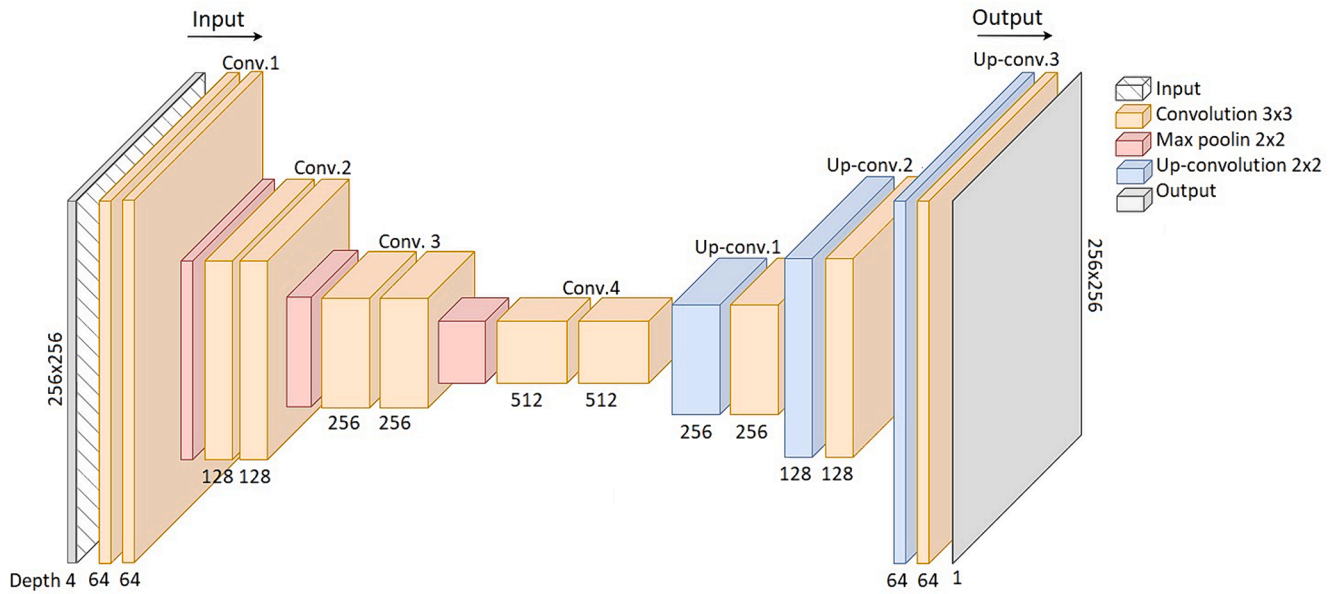


Fig. 5. Two-dimensional U-Net network structure. Used two 3x3 convolutions followed by a max pooling of kernel size 2 which are responsible for down sampling. The numbers below the rectangles indicate the respective channel depth.

$$\text{ReLU}(x) = \max(x, 0) \quad (5)$$

Where x represents the input to the function.

2.2.4.1. Data generation. The input data comprising 226 Sentinel-1 images, meticulously after shuffling, divided into training and testing sets with a split ratio of 70 % for training (158 images) and 30 % (68 images) for testing. Each Sentinel-1 image was further dissected into 999 patches, contributing to the richness and diversity of the dataset. To enhance the dataset's variability and improve the model's robustness, data augmentation techniques such as flipping, and rotation were applied (2 times). This augmentation process effectively doubled the amount of initial data, ensuring a more comprehensive representation of the underlying patterns in the images (Krizhevsky et al., 2012). Therefore, the overall number of U-Net models in the combined training and testing datasets, considering augmentation, was 451,548.

2.2.4.2. Training and testing the model. The encoder comprised a series of convolutional layers (Conv.1 through Conv.4: with 64, 128, 256, and 512 filters, respectively). The increasing number of filters at each layer allows the model to extract features at varying granularities. Starting with basic edge detection, the model progresses to capture more complex patterns like water spread and accumulation, which are critical markers for flood detection. The dimensions of these filters were designed to process a range of spatial data inputs rapidly, ensuring that the model operates swiftly, a necessity for NRT response. In tandem, max pooling layers with a 2×2 window size serves a dual purpose: reducing dimensionality, which accelerates the computational process, and abstracting the essential features that contribute to flood delineation. This selective focus on salient features is crucial for maintaining model performance despite the real-time constraints of data processing and analysis (Kabir et al., 2020; Wang et al., 2020; Hosseiny, 2021).

During the decoding phase, up-convolutional layers, aligned with the encoder's depth, incrementally up sampled the feature maps to the input's original resolution. This precise reconstruction is vital for accurate, pixel-level classification of the flood extent, enabling the delineation of affected areas with high fidelity.

The ReLU activation function was used during the training phase to

facilitate learning. Therefore, the raw output was passed through the activation function, which squashes the value to the range $[0, 1]$. Moving forward, a thresholding step was applied to the continuous probability values. Thresholding used as a decision rule that converted these probabilities into binary classifications based on the chosen threshold (Figure S2). This was the final convolutional operation, which helped in summarizing the information learned by the network and preparing it for the desired output format.

Adam optimization algorithm as a smart helper for training was used globally for the entire neural network during the training process to update the weights (Kingma and Ba, 2014). As per Kingma and Ba (2014), this approach is computationally efficient, requires minimal memory, works well even if the gradient scales change, and is particularly good for handling large-scale problems with a lot of data or parameters. Fig. 6 outlines key details providing a comprehensive overview of this study architecture's configuration. This well-structured U-Net network design holds promise for flood predictions, demonstrating the thoughtful integration of Sentinel-1A imagery and spatial data layers to effectively process, analyze, and understand intricate features or structures within input data.

2.2.4.3. Evaluation metrics. A confusion matrix is a powerful tool in the evaluation of segmentation models, providing a detailed and intuitive summary of the model's performance (Ulmas and Liiv, 2020). It compares the predicted class labels with the true class labels, yielding four key metrics: True Positives or TP (correctly predicted positive instances), True Negatives or TN (correctly predicted negative instances), False Positives or FP (instances incorrectly predicted as positive), and False Negatives or FN (instances incorrectly predicted as negative). This matrix enables the calculation of various performance metrics, such as accuracy, precision, recall, specificity, and the F1 score, offering a more nuanced understanding of the model's effectiveness. Evaluation metrics are defined in the Equations (6) to (9).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (6)$$

$$\text{F1 Score} = \frac{2 (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (7)$$

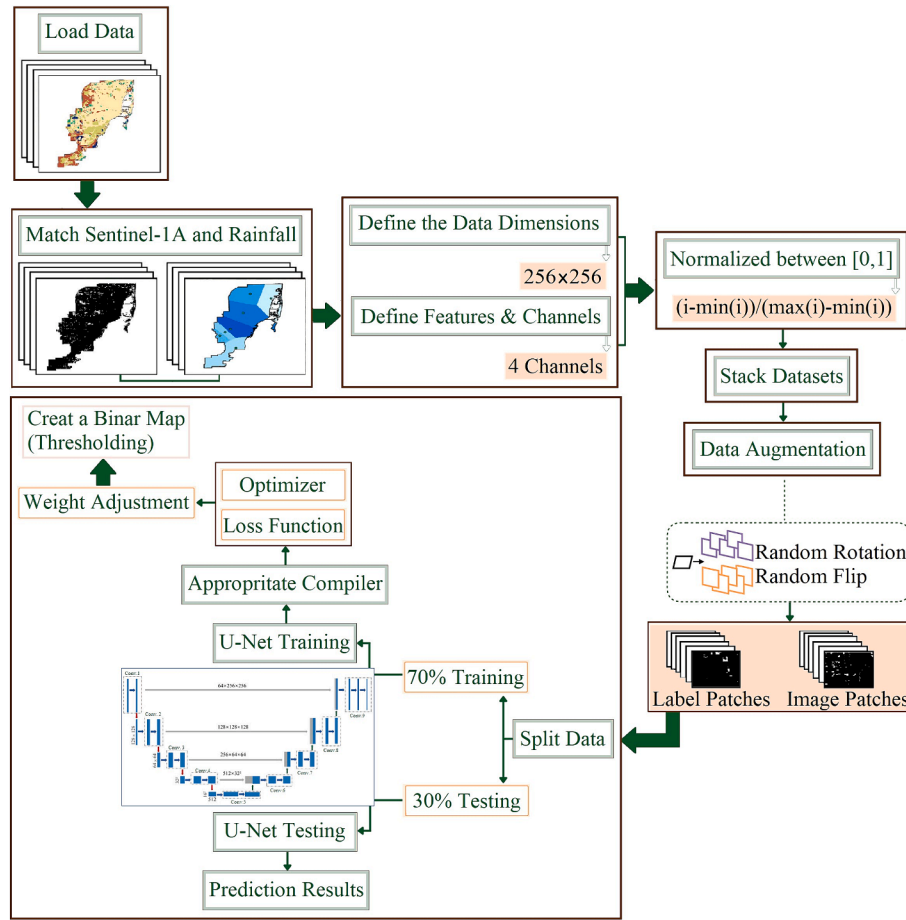


Fig. 6. U-Net schematic for feature extraction and up-sampling matrices for reconstructing intricate data patterns. Pointing to the U-Net architecture modified for flood prediction of 256×256 pixels patches with four channels (slope, HSG, imperviousness, and rainfall).

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

The Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) curve, AUC-ROC score, is another evaluation metric for

overlay analysis was utilized. To gauge the accuracy of predicted flood events corresponding to pixels with historical occurrences, the approach involved counting instances where observed flood events aligned with the predictions. This count was then divided by the total number of observed flood events, yielding the ground truth index (Equation (10)).

$$\text{Ground Truth Index} = \frac{\text{Number of Observed Flood Events corresponding to Historical Flood Events Pixels}}{\text{Total Number of Observed Flood Events}} \quad (10)$$

assessing the performance of classification models, particularly in multi-class classification problems. The AUC-ROC curve is created by plotting TPR against FPR and provides a visual representation of a model's ability to discriminate between positive and negative classes. AUC-ROC score can take any value between 0 and 1. The higher score, the better in distinguishing between different classes, whereas it equals to 0.5 indicates that the model lacks class separation capacity, and its predictions are equivalent to random guessing.

To demonstrate the practical utility of the developed flood prediction model as a spatially and temporally accurate flood mapping tool, an evaluation was conducted using historical flood events. To do so, the historical dataset used for validation included 126 flood reports from online platforms and 50 flood locations from 311 flood records. To assess the precision of the integrated flood vulnerability map, a GIS

2.3. Region of interest (ROI)

The application of this flood prediction model was exemplified within the urban areas of Miami-Dade County, Florida, serving as a case study to demonstrate the model's capabilities in a real-world problem (Fig. 7). While this instance provides a tangible example of the model's utility, the methodology and model's applicability are not confined to this region alone. The ROI was considered as the urban development areas within the Miami-Dade County, which is located in the south-eastern region of Florida, USA (Fig. 7). According to the US Census Bureau, the county spans an area of 6296.26 km^2 (2,431 square miles), with 4915.79 km^2 (1,898 square miles) being land and 1380.46 km^2 (533 square miles) composed of water. It ranks as the third-largest Florida county in terms of land area and the second largest in overall

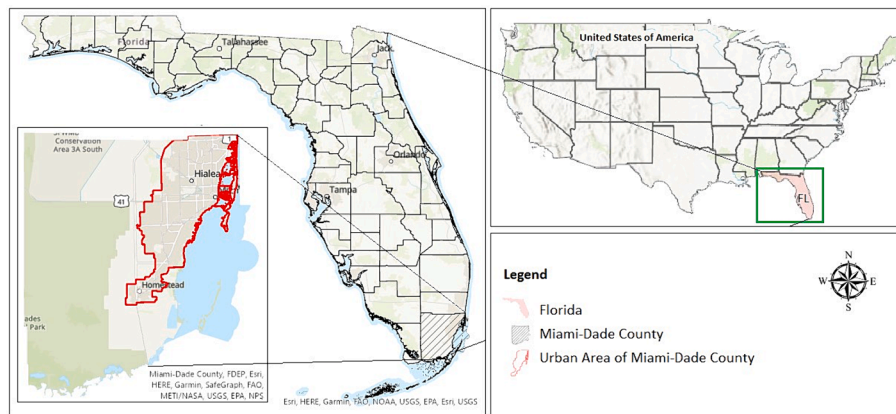


Fig. 7. Location of the ROI: Urban area of Miami-Dade County, Florida, USA.

area (United States Census Bureau, 2011). The county's distinctive location makes it susceptible to flooding caused by heavy rainfall and storm surges. Bordered by significant water bodies such as the Atlantic Ocean, Biscayne Bay, rivers, lakes, and canals, Miami-Dade County faces increased vulnerability to flooding incidents (Miami-Dade County, 2022). The County is naturally a low-lying area vulnerable to urban flooding. The primary focus of this study is on urban flooding in the AOI. The region is prone to flood-related damage due to extreme rainfall, tropical storms, hurricanes, and storm surges (Zhang and Sheng, 2015). Notably, the most significant rainfall events coincide with tropical storms and hurricanes, which also contribute to elevated urban flooding (Sukop et al., 2018). The climate in Miami-Dade County is characterized by distinct wet and dry seasons, with most of the rainfall happening between June and October. The annual precipitation levels vary from 101.4 cm to 151.6 cm (40 to 60 in.) per year, depending on the specific location (Brouillette and Zierden, 2017). However, with its proximity to sea level and shallow underground water reserves, substantial rainfall occasionally lacks proper drainage, resulting in periodic urban flooding across certain county areas (FEMA, 2023).

Moreover, urban development and population growth in the County have led to increased imperviousness as roads and buildings, which means more runoff and flooding during heavy rain events. More than two-thirds of Miami-Dade County's transportation system is exposed to risks from flooding, posing threats to both local communities and the functioning of transportation infrastructure (Chen et al., 2015).

3. Results and discussion

3.1. Flood inventory map

The outcome of applying Otsu thresholding to Sentinel-1 imagery to create a binary flood inventory map is presented here. The data spans a decade, from 2014 to 2023, with the objective of segmenting flooded from non-flooded areas within the region of interest (ROI). Fig. 8 exhibits one of the resultant flood inventory maps dated September 15, 2015. This particular map was chosen as a representative example to demonstrate the effectiveness of the Otsu method, which is used to discern the optimal separation point between the two classes of pixels, flooded and non-flooded. The histogram analysis underpinning this method is based on the premise that there are two distinct peaks representing each class. The Otsu threshold is the valley that lies between these peaks, indicating the point of minimum overlap between the flooded and non-flooded distributions (further details in the Supplementary Document (Figure S2)). By applying this threshold to each year's data, a clear boundary for distinguishing between the two conditions was established, same as Fig. 8. For the U-Net deep learning architecture, these maps were used as labeled data to train the model.

The analysis of the flood inventory map was done to assess the

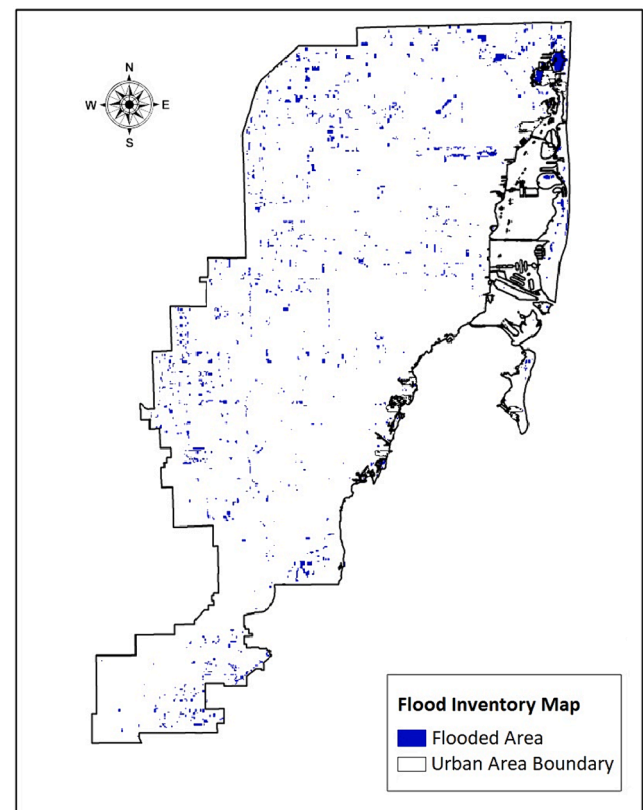


Fig. 8. Representative flood inventory map from Sentinel-1 imagery using Otsu Thresholding. This map, randomly selected for its illustrative value and processed using SNAP, dates from September 15, 2015. It showcases the binary classification of flooded (blue) and non-flooded (white) regions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

accuracy and reliability of flood detection. To do so, 116 recorded flood events were utilized as a benchmark against which the detections from the Otsu method were compared. Achieving a Ground Truth Index of 91.12 % indicates a high level of concordance between the detected flooded areas and the actual events on the ground. The Otsu method for detecting flooded areas achieved more than 90 % agreement or concordance with actual, recorded flood events. Achieving a Ground Truth Index of 91.12 % indicates that the Otsu method demonstrated a high level of accuracy in identifying flooded regions, which has been identified well for its practical utility in disaster management and

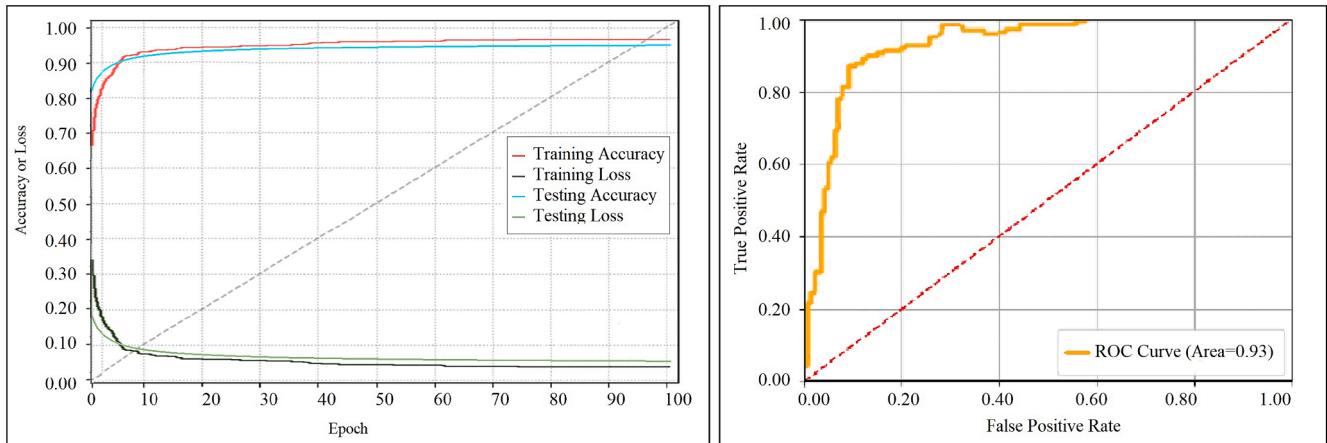


Fig. 9. Performance metrics for U-Net Model. Left: Illustrating changes in loss and accuracy curve across 100 epochs for the entire data. Right: AUC ROC curve, offering a comprehensive evaluation of the U-Net model's performance over the training epochs.

mitigation endeavors (Chen et al., 2021; Markert et al., 2020; Liang and Liu, 2020).

3.2. U-Net flood mapping results and evaluation

Fig. 9 graphically represents the U-Net model's training process, illustrating the accuracy and loss curves over 100 epochs to demonstrate the model's learning dynamics. Notably, the convergence of the training and testing accuracy lines highlights the model's generalization capabilities, which are essential for preventing overfitting. This feature is critical in NRT flood mapping applications, where the reliability of the model on fresh, real-world data is imperative. The rapid increase in accuracy within the initial 10 epochs signifies a swift learning rate and convergence to an optimal solution, minimizing the need for prolonged training (Li et al., 2023). This is further supported by the loss curves, which show a steep decline initially and then stabilize, indicating the model's ability to quickly assimilate new information, a key attribute for systems used in real-time scenarios. Additionally, the ROC curve in Fig. 9 confirms the model's stable learning trajectory and its effective discrimination between flooded and non-flooded classes, with an AUC of 0.93 %. This high AUC value demonstrates the model's ability to maintain a low false positive rate while maximizing the true positive rate. The model's high training precision and recall rates, 97.03 % and 88.62 % respectively, underscore its effectiveness in accurately identifying true positive flood instances and minimizing false positives. The F1 scores (92.49 % in training and 89.85 % in testing) reflect a balanced approach to precision and sensitivity, optimizing the detection of actual flooded areas while avoiding incorrect flood identifications. The overall accuracy rates, 97.05 % in training and 93.73 % in testing, further validate the model's ability to produce reliable flood maps, confirming that its predictions are trustworthy for operational use. The model evaluation results are presented in detail in Table 3.

For illustrative purposes and enhancing the understanding of the model's predictive capabilities, three images (256 × 256) were randomly selected to showcase the model's predictions and evaluation results (Fig. 10). This visual representation compares the flood predictions made by the U-Net model against satellite maps overlaid with ML predictions, effectively situating the predicted flood areas within the actual urban landscape context (see Fig. 10, panels g, h, and i). For three

randomly selected patches within the ROI, flood inventory maps were generated using SNAP software (a, b, and c). These were compared with corresponding flood prediction maps produced by the U-Net model (d, e, and f). Ground truth maps depicting actual flooded areas were also available for comparison (g, h, and i). To facilitate a closer examination, randomly selected zoom-in views (j, k, and n) provide a detailed look at the model's predictions. The random selection of imagery for Fig. 10 illustrates the model's capabilities across diverse scenarios. In panels j, k, and n, the model identifies blue spots, indicating predicted flooding areas.

3.3. Ground truthing

In the ground truthing evaluation process, a random sample year was chosen and integrated with a dataset containing information about spatial flood influencing data and rainfall. The selected year aimed to capture typical flood conditions or those marked by significant flood events. This allows for a comprehensive assessment of the model's performance under varying flood scenarios, contributing to a more nuanced evaluation. This integration involved overlaying 116 validated flood locations onto their corresponding positions on the flood prediction map, the result of the U-Net model. The model demonstrated accurate predictions with a Ground Truth Index of 84.05 %, suggesting that the model's algorithms and features were adept at recognizing flood-associated locations within the provided satellite imagery. This comprehensive approach strengthens the credibility of the model's performance evaluation by aligning it with real-world flood events and conditions. As an example, the outcomes of this ground truthing process for Brickell area, Miami-Dade County on July 7, 2018, year is illustrated in Fig. 11.

3.4. Input data evaluation

To evaluate the performance of the model using various input data, we conducted an analysis using eight different combinations of input data including the original set of input data (Slope + HSG + Imperviousness + Rainfall) in addition to seven other cases as seen in Table 4. Rainfall, as the primary cause of pluvial flooding, is the critical component that enables our model to function in an NRT context and, hence, was considered in all cases. The performance metrics for all the models trained with various combinations of data (slope, HSG, imperviousness, rainfall) are presented in Table 4.

Precision, recall, F-1 score, and overall accuracy for all cases in Table 4 vary within the range of (90.03–94.07), (83.27–86.00), (86.52–89.85), and (91.57–93.73), respectively, indicating a good performance (Hosseiny et al., 2020) for all the combinations of data.

Table 3
Model performance metrics (%).

	Precision	Recall	F1 Score	Overall Accuracy
Training	97.03	88.62	92.49	97.05
Testing	94.07	86.00	89.85	93.73

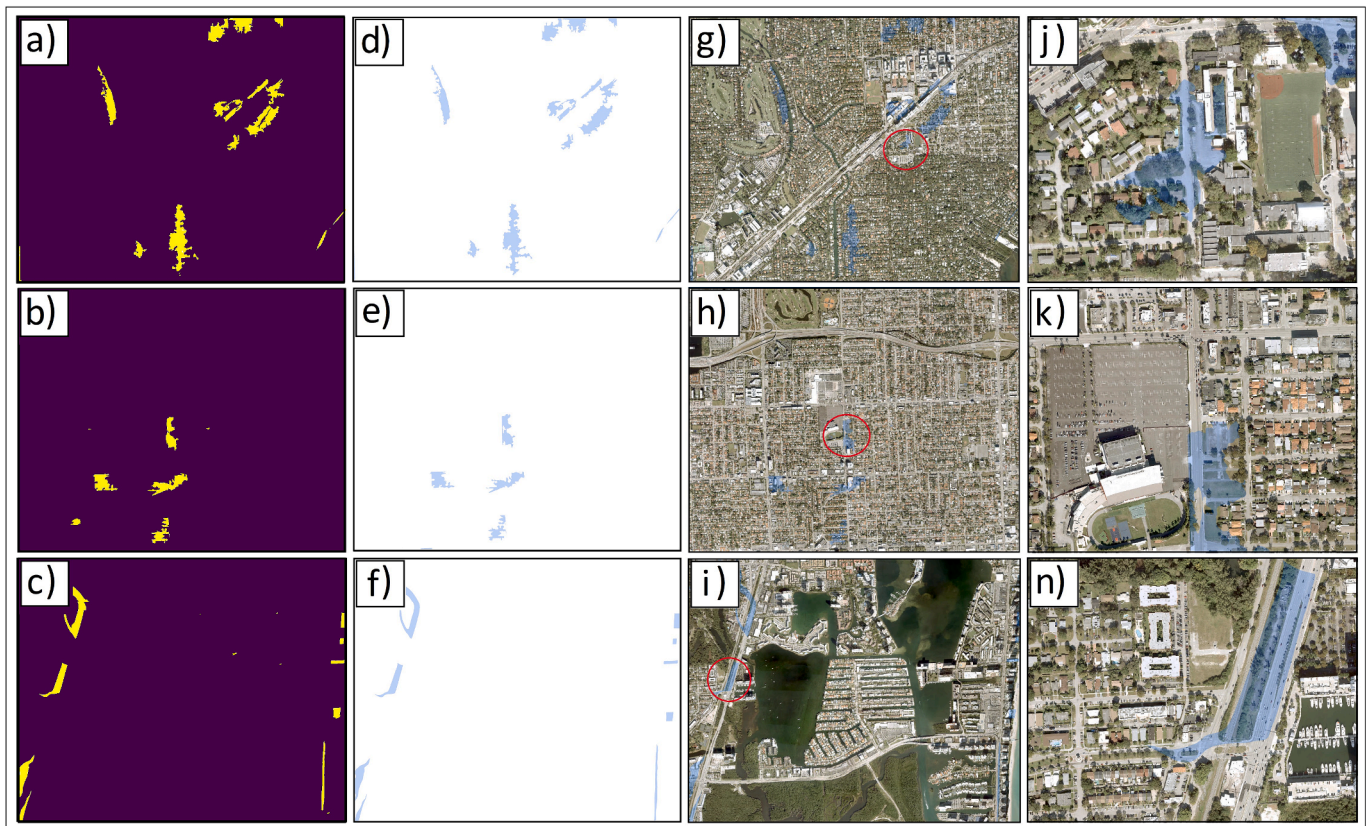


Fig. 10. Visualization of flood detection and verification. Columns represent different stages of analysis for three randomly selected patches within the region of interest (ROI): a, b, and c) flood inventory maps generated by SNAP software for three randomly selected patches within the ROI, presented flood events with yellow highlights; d, e, and f) show the U-Net model's flood prediction maps for the corresponding patches, with predicted flooded areas in blue; g, h, and i) represent satellite maps with overlaid ML predictions, placing the predicted flooded areas within the real-world context of the urban landscape; j, k, and n) randomly selected zoom-in views from the ground truth imagery for a detailed inspection of specific locations predicted as flooded by the model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

However, Table 4 show that the model's performance improves by including more input data (features). This indicates that each input data (feature) has some contributions toward more accurate prediction of flooded areas. It is noteworthy that flood events are the result of complex interactions between various factors (data) and some data can make more contributions to the enhancement of the model performance. The model trained with all data (first row in Table 4) stands out as the best case because it leverages a comprehensive range of input data, resulting in maximum values of precision, recall, F1 Score, and overall accuracy for flood prediction. In addition to the types of input data, the distribution of data over ROI also contributed to model performance. The developed model works based on pixels. Hence, the variation of pixel values, even within a small range, can make changes to the model performance. For instance, the higher variability of slope values in adjacent pixels compared to other data such as HSG, slope data ended up showing up in the first three rows of Table 4. Similar discussion applies to imperviousness that found to be another important factor. Flooding patterns were better recognizable when both slope and imperviousness data were available. Without imperviousness data, the model over-estimated flood extents in areas with infiltration potential, increasing false positives. In general, the models trained with more comprehensive datasets, in terms of both variety and distribution of data, were able to make more accurate predictions.

4. Summary and conclusions

This study aimed to leverage the advances in AI (DL) and remote sensing (satellite imagery) to improve flood mapping prediction

capabilities for addressing the increasingly severe issue of urban flooding, exacerbated by the intensified urban development and climate change impacts, with potential RT/NRT applications. U-Net models, as part of CNNs, can be trained on large datasets to recognize patterns and make predictions. The U-Net architecture was adapted for image segmentation, along with a combination of all the existing Sentinel-1 imagery and rainfall data over the study region within the period of 2014–2023 and other geospatial data (DEM, HSG, and imperviousness) to develop a comprehensive urban flood prediction model. Miami-Dade County, Florida was used as the study region to demonstrate the applicability and performance of the model. This flood prediction model effectively utilized the VV polarization mode of Sentinel-1 imagery to detect and highlight water bodies. Validation and ground truthing against historical flood events from 2014 to 2023, using 311-flood records and online flood reports, ensured the model's accuracy and real-world applicability.

One limitation of this work is the uncertainties related to the correspondence between daily rainfall data and the resulting flooded areas captured from satellite images. The developed model assumed that the flooded areas observed from satellite images were the results of the rainfall that occurred on the same day that the satellite image was taken. Given that Sentinel-1 captures images at specific times of the day once every 6 days, it is essential to consider the dynamics of rainfall distribution. Flooded areas in the satellite images may be as a result of rainfall events that occurred on the same day as the images were captured, the day before, or even earlier. Therefore, to accurately map and predict flooded areas, it is necessary to examine the distribution of rainfall over time before training the U-Net model. This includes identifying which

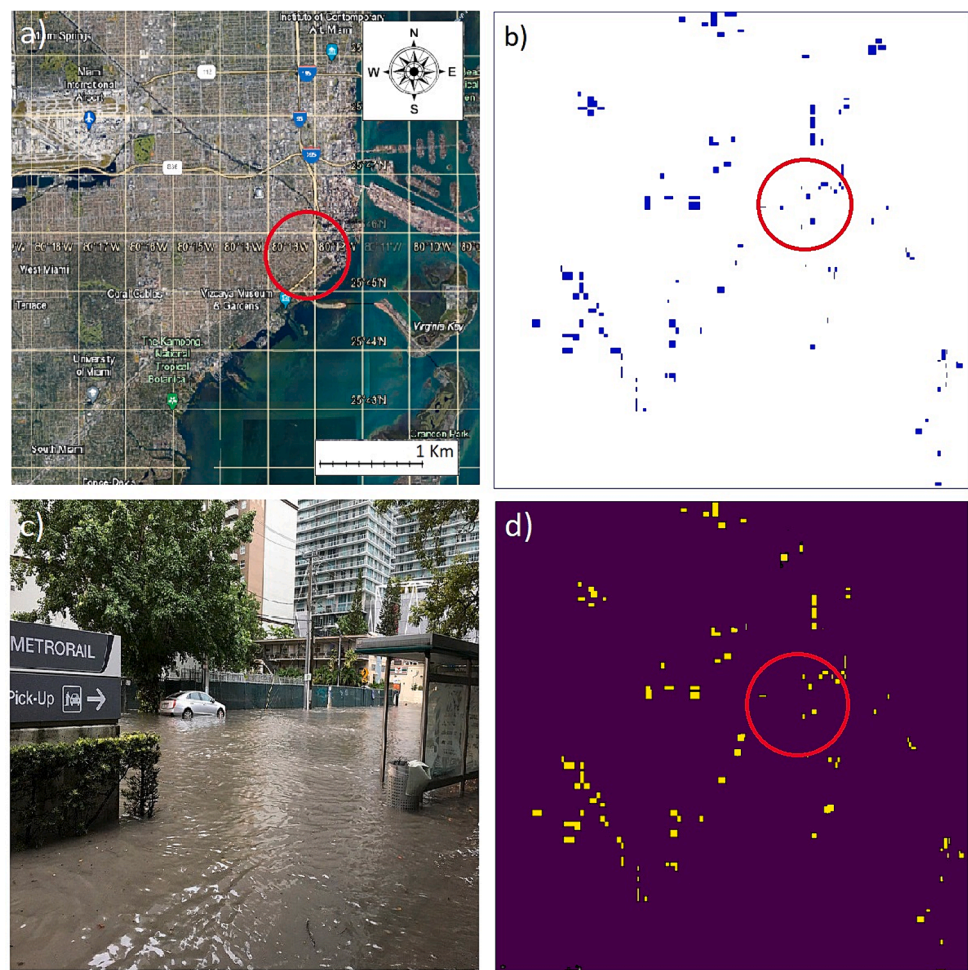


Fig. 11. Detailed flood analysis in Brickell, Miami-Dade County, Florida. a) aerial view from Google Earth of the Brickell area; b) a focused snapshot from the model's flood prediction output, illustrating the delineation of flood extents within the area in blue color, which is then compared to ground-level observations; c) photographs of flooding in Brickell sourced from Twitter user posts, taken on the same date as the model's predictions, serving as real-time verification of flood impact; d) flood inventory maps created using SNAP software, documenting the extent and distribution of flooding in Brickell as a reference for model validation where yellow spots represent flooded areas. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4
Model performance metrics for different sets of input data (%).

Input Data	Precision	Recall	F1 Score	Overall Accuracy
Slope + HSG + Imperviousness + Rainfall	94.07	86.00	89.85	93.73
Slope + Imperviousness + Rainfall	93.47	85.76	89.45	93.35
Slope + HSG + Rainfall	93.22	85.59	89.28	93.18
HSG + Imperviousness + Rainfall	92.85	85.34	88.97	92.90
Imperviousness + Rainfall	92.14	84.97	88.42	92.75
HSG + Rainfall	91.93	84.42	88.05	92.60
Slope + Rainfall	91.67	84.25	87.82	92.45
Rainfall	90.03	83.27	86.52	91.57

rainfall events are directly related to the flooded areas observed in the satellite images and distinguishing them from other rainfall occurrences that may not contribute to the flooding. Hourly monitoring of weather stations can be helpful for understanding the temporal correlation between Sentinel-1 imagery and short-term rainfall patterns, enhancing the accuracy of flood mapping and predictions. Despite this acknowledged limitation, the use of daily rainfall data in this study offered a significant reduction in computational burden while providing adequate accuracy for NRT applications. The focus on quick computations aligns

with the need for timely decision-making in urban areas susceptible to sudden and intense rainfall. It recognizes the existing gap in RT/NRT flood prediction, where quick predictions close to the present time are essential but must also maintain an adequate level of accuracy for decision-making.

By using RT or NRT rainfall data (e.g., rainfall data during or shortly after storm events), the model can be used as part of a RT or NRT system to provide rapid and accurate flood extent predictions, offering crucial insights for timely responses in urban areas. As new Sentinel-1 images become available, they can be incorporated into the DL model to extend the training dataset. Similarly, updated versions of the other flood influencing data (e.g., DEM and imperviousness) can be used for model development, as they become available, maintaining the model as a live tool in the future. Potential applications of the model include, but are not limited to, informing evacuation planning, guiding effective infrastructure development, operation, and maintenance strategies, and implementing targeted flood mitigation measures in urban areas.

CRedit authorship contribution statement

Nasim Mohamadiazar: Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ali Ebrahimian:** Conceptualization, Supervision, Funding acquisition, Methodology, Validation,

Writing – original draft, Writing – review & editing. **Hossein Hosseiny:** Methodology, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The links to data are provided in the paper.

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Appendix A. Supplementary data

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